

KARTAVYA KOTHARI

MACHINE LEARNING SCIENTIST • SPORTS VIDEO AI • RECOMMENDATION & EXPERIMENTATION

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Applied ML scientist with five years building recommendation, experimentation, and model systems at 30M+ MAU scale.
Now focused on sports video understanding: tracks, events, highlights, insights, and player-level signals from raw footage.

EXPERIENCE

PlayVisionAI • Sports video AI startup, Mumbai, India Aug 2026 - Present

Principal Machine Learning Scientist

- Build video primitives that convert sports footage into player and object tracks, events, trajectories, highlights, match insights, and sports-science signals.
- Develop and evaluate representations using JEPA/V-JEPA 2, VideoMAE, and vision transformers; use Qwen-family VLMs for multimodal extraction, fine-tuning, and validation.
- Design video decoding, training, and inference across CPUs, NVIDIA GPUs, and GCP TPUs; profile backends, batching, and accelerator utilization.

Dream Sports • Mumbai, India Jun 2021 - Jul 2026

Lead ML Scientist - Dream Play

Nov 2025 - Jul 2026

- Built ML for a global padel product, converting match footage into player ratings and personalized technical, tactical, and physical feedback.
- With sports scientists, formulated the skill score from 60+ detected metrics and translated model outputs into player feedback.
- Designed distributed Ray pipelines for computer vision, GenAI insights, highlights, and leaderboards from research through launch.

Dream11 - Data Scientist; promoted to Senior ML Scientist, then Lead ML Scientist Jun 2021 - Oct 2025

- Owned gameplay and lifecycle personalization and experimentation for 30M+ MAU and approximately 15M peak CCU.
- Built the WiDIR contest recommender and smart-discounting systems; productionized neural ranking for product bundling and promotions.
- Built FENCE for real-time multiple-account detection; published the system at AI-ML Systems 2023.
- Designed distributed GPU training and inference on Ray and Databricks; standardized reusable model pipelines.
- Ran dynamic and static-cohort experiments, including network effects; established evaluation guardrails.

IIT Bombay • Mumbai, India Sep 2019 - Jan 2021

Research Assistant - Prof. Soumen Chakrabarti

- Designed entity-alignment methods for heterogeneous and multilingual knowledge graphs including Wikidata and DBpedia, surveying approaches through BERT-based models.
- Developed a target answer-type predictor using contextualized question embeddings that outperformed earlier published results.

PUBLICATIONS

Personalized Contest Recommendation in Fantasy Sports ECML-PKDD 2025

Co-authored and presented WiDIR, a production-scale contest recommender, at ECML-PKDD in Porto. arXiv:2508.14065

Efficient Feature Interactions with Transformers: User Spending Propensity in Gaming ACM KDD 2024

Co-authored and presented transformer-based feature-interaction work at ACM KDD in Barcelona.

FENCE: Fairplay Ensuring Network Chain Entity for Real-Time Multiple ID Detection AI-ML Systems 2023

Co-authored and published a production fraud-detection system for fantasy sports. arXiv:2310.05651

EDUCATION & RECOGNITION

M.Tech, Computer Science 2019 - 2021

Indian Institute of Technology Bombay
Machine Learning and NLP; thesis on multilingual knowledge graphs and corpus linkages.

B.E., Computer Science 2015 - 2019

Vidyalankar Institute of Technology, Mumbai
Computer science engineering.

Recognition

- GATE 2019 - All India Rank 45
- Google Research India AI Summer School 2020
- ET Campus Stars - Top 100 of 37,000

TECHNICAL SKILLS

Research

Sports video understanding; representation and multimodal learning; tracking; experimentation and model evaluation

Models

JEPA/V-JEPA 2; VideoMAE; CNNs; vision transformers; Qwen-family VLMs; vision-language/action and world-model approaches; ranking and recommendation

Systems

PyTorch; Ray; Databricks; Spark; NVIDIA GPUs; GCP TPUs; distributed video training and inference

Languages

Python, C++, SQL